2+5

Nepal Engineering College

Level: Bachelor — Year:2026
Program: BE — Full Marks: 100
Course: Embedded Systems — Pass Marks:45
Time: 3 Hours

Candidates are required to give their answers in their answer in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions

1	a)	What is an embedded system? List and define the three main characteristics of embedded system that distinguish such system from other computing system?	8
	b)	Describe the architecture of the AVR (ATmega32) microcontroller with the help of a neatly labeled block diagram and highlight the key features of its architecture.	7
2	a)	Write a program to toggle the LEDs connected to PORTD of the ATmega32 microcontroller Also, include a properly labeled diagram illustrating the connection of LEDs to the GPIO pins of PORTD.	8
	b)	Explain the concepts of UART. Write a program to display "Hello World" on UART with default configuration.	7
3	a)	Explain the issues that arise during resource sharing in a multitasking environment and How such problem can be addressed in RTOS.	8
	b)	Explain priority inversion, Discuss the methods to address this problem.	7
4	a)	Describe various VHDL modeling styles with examples and illustrate how a full adder can be modeled using structural modeling style.	8
	b)	What is a VHDL test bench, and write a VHDL program to implement a 4:1 multiplexer.	7
5	a)	What is communication protocol? Draw and explain the frame structure of a UART packet.	8
	b)	Write a program to interface an LED with a microcontroller such that it blinks ON and OFF using Timer0 in Normal mode. Generate a delay of 1 second using the hardware timer and show the complete delay calculation. Include a suitable circuit diagram.	7

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6 a) Write a program to interface an LCD with a microcontroller in 8-bit mode to display the message "Gen-Z Revolution". Include a suitable circuit diagram showing connections between the LCD and microcontroller? 7

b) What is the Internet of Things (IoT), and Explain IoT architecture with its main components? 8

7 **Write shorts notes on: (any two)** 5+5

a) Timer and Counters of AVR microcontroller

b) Task Scheduling Algorithms in RTOS

c) VHDL and Application of VHDL

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Madan Bhandari College of Engineering
Urlabari-3, Morang
Final Assessment- 2026, Fall

Level: Bachelor
Programme: BE(Computer) **Full Marks : 100**
Course: Embedded System (New) **Pass Marks : 45**
Year/Part:III/I **Set A** **Time : 3 hrs.**

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

1. a) Define an **Embedded System**. Explain how embedded systems differ from **general-purpose computing systems**. Discuss any **three applications** of embedded systems with examples. [7]

OR

Explain different components of ES with a figure. Prove that **ATM**, is an example of ES.

b) Design an embedded system for an **Industrial Temperature MonitoringSystem**. [8]
Discuss:
   a) System architecture
   b) Design metrics and constraints
   c) Hardware and software components
   d) Challenges and design steps

2. a) Explain the key features of the AVR(Atmega16/32) architecture with a figure and also include the role of General-Purpose Registers,ALU, Program Counter, and Watchdog Timer in AVR.[8]

b) Write a simple Embedded C program to blink an LED connected to PORTB pin 0 of an AVR microcontroller. [7]

OR

Write a program to store and retrieve a value from the EEPROM memory of an AVR microcontroller. [7]

3. a) Define Explain **RTOS. Explain types of Real time System.** How does an RTOS differ from a General Purpose Operating System (GPOS)? [7]

OR

Describe key features of RTOS and compare **FreeRTOS and VxWorks**.

b) Explain **task synchronization mechanisms** used in RTOS. How are **deadlock and priority inversion** handled in real-time systems?[8]

OR

Explain **priority inversion** with an example. Discuss techniques used to overcome this problem in RTOS. [8]

4. a) What is a **VHDL test bench**? Write only the test bench code for a **2:1 multiplexer**. [7]

b) Explain different **VHDL modeling styles**. Demonstrate how a **4:2 Encoder** can be implemented using **behavioral modeling**. Verify it by drawing wave.[4+4]

OR

Explain different types of data types in VHDL. VHDL code for Combinational circuit like 4:2 **Full Adder** by using behavioral modeling style. Also Verify it by drawing wave.[4+4]

5. a) Design an embedded system for Wearable Health Monitoring devices.. Discuss how UART, SPI, and I²C interact with sensors, displays, and controllers. [8]

b) Explain the working of **GSM/GPRS module** in embedded systems. Define **LoRa** and compare it with **Wi-Fi and Bluetooth.**[7]

6. a) Write a program to interface a **seven-segment display** with a microcontroller to display digits **0–9** sequentially with a suitable delay. Draw a neat circuit diagram. [7]

b) A **Smart Parking System** uses IoT devices to update parking availability. Answer the following:
i. Explain MQTT topic design for multiple parking slots
ii. How to handle server failures?
iii. Explain system operation using Arduino and sensors
iv. List communication protocols for cloud connectivity
v. Methods for real-time monitoring enhancement

7. **Write short notes on (Any two): *(2 × 5 = 10)***
a) Sensor Interfacing b) Multitasking Vs Multithreading
c) Context Switching d) Raspberry Pi Vs ESP32 for IOT

*****All The Best"""""

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Madan Bhandari College of Engineering
Urlabari-3, Morang
Final Assessment- 2025, Fall

Level: Bachelor
Programme: BE(Computer) **Full Marks : 100**
Course: Embedded System (New) **Pass Marks : 45**
Year/Part:III/I **Set-B** **Time : 3 hrs.**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

a) What is an **Embedded System?** List and define the three main characteristics of Embedded system that distinguish such system from other computing systems. [7]

OR

Explain different components of ES with a figure. Prove that **CCTV Camera,** is an example of ES. [7]

b) Design an embedded system for an **Home security system.** [8]
Discuss:

i. System architecture
ii. Design metrics and constraints
iii. Hardware and software components
iv. Challenges and design steps

1. **a)** Explain the key features of the AVR(Atmega16/32) architecture with a figure and also include the role of General-Purpose Registers,ALU, Program Counter, and Watchdog Timer in AVR.[8]

b) Write an AVR program in C to read a digital input from a switch and turn on an LED. [7]

OR

Write an AVR program in C to generate a square wave using Timer 0 and Timer 1 AVR microcontroller. [7]

2. a) How does RTOS differ from a general-purpose operating system (GPOS)in an Embedded system? Explain with key features of an RTOS? Also, differentiate between FreeRTOS and VxWorks.

b) Explain **task synchronization mechanisms** used in RTOS. How are **deadlock and priority inversion** handled in real-time systems?[8]

OR

Explain **priority inversion** with an example. Discuss techniques used to overcome this problem in RTOS. [8]

4. **a)** What is a **VHDL test bench**? Write only the VHDL **test bench code** for a **4:1 multiplexer.** [7]

b) Explain different **VHDL modeling styles.** Demonstrate how a Full Adder can be modeled using **structural modeling style.** Verify it by drawing wave.[4+4]

OR

Explain different types of data types in VHDL. VHDL code for Combinational circuit like **4:2 Encoder** by using behavioral modeling style. Also Verify it by drawing wave.[4+4]

5. a) Design an embedded system for Wearable Health Monitoring devices.. Discuss how UART, SPI, and I²C interact with sensors, displays, and controllers. [8]

b) What is LoRa, and how does it differ from other wireless communication protocols? How does LoRa achieve low power consumption in embedded applications? [7]

6. a) Write a program to interface a seven-segment display with a microcontroller to display digits 0–9 sequentially with a suitable delay. Draw a neat circuit diagram. [7]

b) Industrial Automation System [8]

i. Discuss how Raspberry Pi is used in industrial automation applications.
ii. What protocols (e.g., Modbus, MQTT) are suitable for communication in factories?
iii. How is real-time data logging achieved using Raspberry Pi?
iv. Explain how remote monitoring is enabled through Raspberry Pi's network capabilities

7. Write short notes on (Any two): *(2 × 5 = 10)*

a) Sensor Interfacing b) Multitasking Vs Multithreading
c) Context Switching d) Arduino Vs ESP32 for IOT

*****All The Best"""""

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UNIVERSAL ENGINEERING AND SCIENCE COLLEGE

Level: Bachelor | Semester: 5th computer | Year: 2025
Program: BE | Full Marks: 100
Course: Embedded System(New) | Pass Marks: 45
Time: 3 hrs.

Attempt all the questions.

1. a) Differentiate between embedded systems and general-purpose systems. 7

   b) Explain the applications and domains of an embedded system 8

2. a) Explain the AVR microcontroller architecture with a neat block diagram. 7

   b) What are the main differences between standard C and Embedded C. What is Data types in Embedded C? 8

3. a) Define resource sharing in RTOS. What is deadlock? Explain the four necessary conditions for deadlock and describe methods to prevent or avoid deadlock in real-time systems. 8

   b) What is context switching in an RTOS, and what impact does it have on system performance and response time? 7

4. a) Explain different modeling styles in VHDL. Write VHDL code for a 4:1 multiplexer using behavioral modeling. 7

   b) Write VHDL code for a Half Adder using behavioral modeling. Explain the data types used and draw the simulation waveform. 8

5. a) Discuss various wireless communication technologies used in IoT and embedded systems. 7

   b) Write an embedded C program to interface a 16x2 LCD with a microcontroller and display the message "EMBEDDED SYSTEM". Explain the code. 8

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6. a) Discuss actuator interfacing techniques in embedded systems. Explain the working and interfacing of: 8
   - DC motor
   - Stepper motor
   - Servo motor

   Compare their characteristics and applications.

   b. What is an MQTT client? Write the steps involved in establishing an MQTT connection between an IoT device and a broker. 7

7. Write short notes on: **(Any two)** 2×5
   a) USART
   b) I2C
   c) Raspberry Pi

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UNITED TECHNICAL COLLEGE

Semester-Fall

Level: Bachelor
Programme: BE
Course: Embedded system

Year: 2025
FullMarks: 50
PassMarks: 23
Time: 1.5hrs

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt any 3 question from 1 to 6 question number 7 is compulsory.

1. a) What is Embedded system? Write any six difference between general purpose system and Embedded system. 2+6=8
   b) Explain the types of embedded system based on performance of microcontroller? Also write some application of embedded system. 3+4=7

2. a) What is RTOS? Explain soft and hard real time system. 7
   b) What is task scheduling? Explain different type of task scheduling algorithm. 8

3. a) What is VHDL? Explain why VHDL is used for designing. 8
   b) Write a VHDL code for 4 to 1 MUX. 7

4. a) Explain serial and parallel communication with suitable example. Compare USART I2C and SPI. 3+5=8
   b) Explain TCP/IP based on embedded system. 7

5. a) Explain IOT in detail with suitable example. 8

   b) Explain MQTT protocol with suitable block diagram. 7

6. a) What is sensor? Explain analog and digital sensor with suitable example. 7
   b) What are different displaying interfacing used in embedded system? Explain. 8

7. Write short note on *(Any One)* 10
   - Wireless communication
   - Architecture of AVR
   - Multithreading in RTOS

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test I

Date: 2082/09/27			
Level	BE	Full Marks	50
Programme	BCT	Time	
Semester	V	1.5 hrs	

Subject: - Embedded Systems

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt All questions.*
- ✓ *The figures in the margin indicate Full Marks.*
- ✓ *Assume suitable data if necessary.*

1. A small irrigation system uses a soil-moisture sensor, a microcontroller and a water pump. The system should control water pump based on moisture level.
   a. Identify the key components of this embedded system. [2]
   b. Sketch the basic block diagram of this system. [2]
   c. Explain the role of the microcontroller in this system. [2]
   d. Why is this system called Embedded, instead of general-purpose computer. [2]

2. Which VHDL modelling style do you think would be used in large scale product design project and why? Write VHDL code for 4-bit binary adder using structural modelling style. [2+5]

3. Write an embedded C program for AVR microcontroller to generate a square wave of 8 microseconds on PB5 using timer0 overflow-interrupt while at the same time data is being transferred from PORTC to PORTD (Assume F =8 MHz). [8]

4. Explain I2C protocol in detail. [7]

5. Write the program to toggle the LEDs connected to PORTD of the Atmega32 microcontroller with 1 second delay continuously. Show connection diagram and for delay you can use predefined delay functions. [8]

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6. Write VHDL code to implement 2-bits 4x1 MUX using behavioral modelling. [7]

   OR

   What is Testbench? Write a VHDL code for simple 2x1 MUX and the testbench code for it.

7. Write short notes on (any two): [2x2.5=5]
   a. Zigbee
   b. LoRa
   c. GSM/GPRS
   d. USART

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date	2082 11 12		
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	V		1.5 hrs

Subject: Embedded System

- *Candidates are required to give their answers in their own words as far as practicable*
- ***Attempt All questions.***
- *The figures in the margin indicate* ***Full Marks***
- *Assume suitable data if necessary*

1. Explain MQTT protocol in IOT. Explain how you use ESP32, Arduino and Raspberry Pi to setup an IOT network. **[8]**

2. Explain the interfacing of LM35 (analog temperature sensor) to AVR microcontroller with code. **[7]**

3. What is PWM? How can you control the width of PWM signals in AVR to control the brightness of a LED? **[8]**

4. Explain about packages and sub-packages in VHDL. **[7]**

5. What is process in VHDL? Write VHDL code for designing 4:1 Multiplexer using behavioral modelling style. **[8]**

6. Explain priority inversion of Tasks in RTOS. How can this be resolved? **[7]**

7. Write Short Notes on anyone: **[5*1=5]**

a. RTOS task states
b. FreeRTOS
c. Half and Full Step Excitation sequence of Stepper Motor

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Pokhara University

Everest Engineering College

Final Internal Assessment
Fall- 2025

Level: Bachelor | **F.M.** 100

Program: BE CMP(5th Semester) | **P.M.** 45

Section: A | **Time:** 3 hrs.

Subject: Embedded System

Attempt all the questions.

1	a)	Draw the internal architecture of AVR microcontroller with its features.	7
	b)	Define embedded system? List down the characteristics of embedded system and explain the three salient features.	8
2	a)	What is the difference between counter and timers in AVR? Write an ALP in AVR to illustrate the use of timer to generate a square wave of 50% duty cycle using polling.	8
	b)	How serial communication is achieved in AVR microcontroller? Write a program in ATmega32 to receive bytes of data serially and out them on Port B. Set the baud rate of 9600, 8 – bit data and 1 stop bit. Use interrupt instead of polling.	7
3	a)	Define Real Time System. Differentiate between Real Time Operating System and General-Purpose Operating System.	8
	b)	Explain Task Scheduling process in RTOS.	7
4	a)	Discuss semaphore based and mutex based resource sharing techniques in RTOS.	7
	b)	Define VHDL. Explain sub program and packages in VHDL with examples.	8

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5 a) Write a VHDL code for 4 x 1 multiplexer using structural modelling style. 7

b) Write a VHDL code for a simple FSM using either Mealy or Moore Model. 8

6 a) Write down a program to interface a temperature sensor with ATmega32 microcontroller. 7

b) Compare and contrast the features of Arduino, ESP32 and Raspberry Pi. 8

7 Write short notes on: (Any two) 2*5=10

a) FreeRTOS vs VxWorks

b) Wireless Communication in Embedded System

c) MQQT

Best Wishes

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POKHARA ENGINEERING COLLEGE
Final Assessment

Level: Bachelor | Semester – Fall | Year :2025
Programme: Computer | Full Marks: 100
Course: Embedded System | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is embedded system? List and define the three main characteristics of embedded system that distinguish embedded system from other computing systems. 2+6

   b) What are the application areas of Embedded Systems. Suppose you are tasked with designing an embedded system for a **"Home Security System"** that include motion detection, door sensor and alarms. Discuss the design merits, constraints, structure, challenges and steps in the design process for developing this system. 2+5

2. a) Draw and explain the AVR microcontroller architecture with its key components. 3+5

   b) Write a program in AVR to store and retrieve a value from EEPROM memory. 7

   OR

   Write a program to turn ON the LEDs connected to PORT D of the ATmega32 microcontroller. Also, include a properly labelled diagram illustrating the connection of LEDs to the GPIO pins of PORT D.

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3. a) What is RTOS? What are the characteristics of RTOS? How does it benefit embedded system applications? 2+3+3

   b) What is context switching? Describe the steps involved in context switching in an RTOS. 2+5

4. a) Differentiate between structural and behavioural modelling styles in VHDL. Explain briefly with examples. 2+6

   b) Write VHDL code to simulate 4 bit binary adder. 7

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a) How would you send an SMS using GSM module in an Embedded application? Define LoRa and how does it differ from other wireless communication protocols. 1+6

b) What is TCP/IP? What are the key layers of the TCP/IP model? Briefly describe their functions. 1+7

a) Write a program to interface a seven-segment display with a microcontroller to sequentially display digits from 0 to 9 with a 2 ms delay. Include a suitable circuit diagram. 7

b) What is MQTT? What are the key components of MQTT protocol. Describe its working principle 1+3+4

Write short notes on **any two:** 2×5

a) Sensors and Actuators

b) Deadlock in RTOS

c) Arduino and Raspberry for IOT

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POKHARA UNIVERSITY

Level: Bachelor | Internal Exam | Year : 2025
Programme: BE Computer (III/I) | Full Marks: 100
Course: Embedded Systems | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1.	a)	Define an embedded system. Explain the key characteristics of embedded systems with suitable examples from automotive or healthcare applications.	7
	b)	Explain the AVR microcontroller architecture. Describe the role of CPU, memory organization, I/O ports, and timers in embedded applications.	8
2.	a)	Explain interrupt handling in AVR microcontrollers. Differentiate between polling and interrupt-driven I/O with suitable examples.	8
	b)	Write a program to store and retrieve a value from the EEPROM memory of an AVR microcontroller.	7
3.	a)	Define Real-Time Operating System (RTOS). Explain hard real-time and soft real-time systems with examples.	8
	b)	Explain task scheduling in RTOS. Compare preemptive and cooperative scheduling techniques.	7
4.	a)	Explain different modelling styles in VHDL. Give one simple example for each.	8
	b)	Write a VHDL program to implement a 4×1 multiplexer and explain its operation.	7
5.	a)	Design a Moore finite state machine for a traffic light controller. Explain the states and transitions clearly.	7
	b)	An IoT-based environmental monitoring system uses multiple sensors connected to an embedded controller. The collected data is transmitted from the controller to a remote monitoring station for analysis.	8
		i. Explain how communication protocols such as UART, SPI, and I2C can be used for communication between sensors and the embedded controller. Compare them in terms of number of communication lines, data transfer speed, and typical	

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applications in embedded systems.

ii. After data collection, the embedded controller sends data to a remote monitoring station. Explain why wireless communication technologies such as ZigBee or LoRa may be preferred over wired communication in this stage. Give at least two practical reasons.

6. a) Write an AVR C program to interface a temperature sensor with an AVR microcontroller and display the measured value on a 16×2 LCD. Clearly explain initialization steps. 7

   b) Compare Arduino, ESP32, and Raspberry Pi as embedded and IoT development platforms in terms of processing capability, power consumption, and application domains. 8

7. Write short notes any two: 2×5

   a) Sensor and actuator interfacing in embedded systems

   b) MQTT protocol

   c) Power optimization techniques in embedded systems

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National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor | Semester : V_Fall | Year : 2025

Program: B.E. Computer | F.M. : 100

Course: Embedded System | P. M. : 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as for as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1.	a)	Compare Embedded systems with general purpose computing systems in terms of design goals and constraints.	7
	b)	Describe the AVR Architecture. Write an embedded C program to read data from PORT B and, if the value is greater than 100 send it to PORT C, otherwise clear PORT C.	8
2.	a)	Discuss the various Serial Communication features available in AVR microcontrollers.	8
	b)	Explain the functionality of counters in AVR. How can you configure a timer to count external pulses? Assume necessary register details.	7
3.	a)	Define Task Scheduling and context switching in the context of a Real Time Operating System(RTOS).	8
	b)	What is priority Inversion? Explain how it can lead to system failure and discuss one method to prevent it.	7
4.	a)	Mention the features of VHDL and the advantages associated with them.	8
	b)	Write a VHDL test bench program for full adder.	7
5.	a)	Write a VHDL program to detect a sequence '1011'.	7
	b)	Write a C program for ATmega32 to display "Hello World" in a 16*2 LCD (4-bit mode).	8
6.	a)	Compare the features and typical use cases of Bluetooth, ZigBee and LoRa wireless protocols.	8
	b)	Write about MQTT protocol used in IoT Communication.	7
7.	Write short notes on any two:		2*5
	a)	Raspberry Pi	
	b)	Deadlock in RTQS	
	c)	Sensor	

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Gandaki College of Engineering and Science

Level: Bachelor | Semester – Fall | Year: 2025
Programme: BE CE V | Full Marks: 100
Course: Embedded System | Time: 3 hrs

Candidates are required to give their answer in their words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions

1. a) What is an Embedded System? List and define the three main characteristics of embedded systems that distinguish embedded system from other computing systems. 8

   b) What is optimization? Explain the different optimization opportunities. 7

2. a) Explain input and output ports interfacing on AVR. 8

   b) What are the key factors a programmer needs to consider when choosing a general-purpose controller? 7

3. a) How does Real time Operating System(RTOS) differ from a general-purpose operating system(GPOS) in an Embedded system? Explain the key features of an RTOS? Also, differentiate between FreeRTOS and VxWorks. 7

   b) Define scheduling? Explain various types of task scheduling techniques in RTOS. 8

4. a) Write a VHDL code to simulate 4 x 1 Mux. 7

   b) Give the VHDL implementation of full adder using both behavioral and structural architecture. And then point out the difference between the two architectures. 8

5. a) Explain UART, SPI and I2C. 8

OR

A company is designing a smart home automation system where various sensors (temperature, motion, gas, etc) and actuators (light, fans, door, locks) communicate for different devices.

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a. Identify and justify suitable communication protocols (wired and wireless) for this system.
b. Compare SPI, I2C, and UART in terms of data rate, complexity, and power consumption for sensor communication.

b) What is TCP/IP? Explain the functions of key layers. 7

6. a) Explain sensor interfacing for analog and digital sensors with examples. 7

b) Explain IOT communication Protocol. Describe the working principle of the MQTT protocol and explain its key components. OR 8

Write an AVR C program to interface a 16x2 LCD (in 4-bit mode) with an AVR microcontroller (e.g. ATmega16/ATmega32). The should:
a. Initialize the LCD in 4-bit mode.
b. Display the message "AVR LCD TEST".
Assumption (Use an 8MHz clock frequency for the microcontroller, include appropriate delays where required).

7. Write short notes on (any two). 2*5

a) LoRa WAN and ZigBee
b) Arduino and Raspberry for IOT
c) Context Switching

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NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Level: Bachelor	FALL-2025	Year: 2025
Program: BE -CE		Full Marks: 100
Course: Embedded System (New)		Pass Marks: 45
		Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable
The figures in the margin indicate full marks
Attempt all the questions.

1. a) List and define the three main characteristics of ES that distinguish ES from other computing systems. Prove that an "Anti-braking system (ABS)" is an example of ES with the necessary components and figure. 7

   b) Imagine a smart traffic management system is being deployed in a city to continuously monitor vehicle flow, traffic density, and signal timing in real time. 8

   i. Compare embedded systems with general-purpose systems in the context of this traffic monitoring and control application.

   ii. Discuss the trade-offs between flexibility, power consumption, real-time response, reliability, and deployment cost.

   iii. Suggest design considerations that make an embedded system more suitable for mission-critical traffic control and safety operations.

2. a) Explain the key features of the AVR architecture with a figure, and also include the role of General-Purpose Registers, ALU, Program Counter and Watchdog Timer in AVR. 8

   b) Discuss how memory is managed in Embedded C for AVR microcontrollers. Explain with C code and a circuit diagram. 7

   Or

   Write an AVR program in C to generate a square wave using Timer 0 and Timer 1 on the AVR microcontroller also mention circuit diagram

3. a) Explain the architecture of FreeRTOS and VxRTOS with their differences (Any 6). 7

   Or

   A modern industrial automation controller must coordinate sensors, actuators, and safety mechanisms under strict timing constraints.

   i. Explain why a real-time operating environment is important in such an embedded control system.

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ii. Compare how task scheduling and timing guarantees differ between a real-time system and a conventional operating system. 8

iii. Describe the essential characteristics required for deterministic and reliable operation in time-critical embedded applications.

b) Smart Factory Robotic Arm Control Tasks: Grip force adjustment every 8 ms, conveyor belt coordination every 40 ms, production log update every 250 ms

Questions:

i. Propose an appropriate scheduling approach for these tasks and justify your choice based on timing requirements.

ii. Describe what happens inside the system when a high-priority grip control task interrupts a currently executing low-priority logging task.

iii. If grip control and conveyor coordination tasks access shared actuator status data, which synchronization technique would you apply? Justify your reasoning.

a) Mention different data operators in VHDL. Write VHDL code for a sequence detector that detects "1101" using behavioral modeling style. 8

Or

What is a VHDL subprogram and a package? Explain the different styles of modeling in VHDL, providing proper examples and highlighting their differences.

b) What is a test bench in VHDL? Write only the test bench code for the Que 4(a). Also, verify it by drawing a wave.

a) A wearable health monitoring device collects data from multiple sensors and transmits the information to a mobile application using both wired debugging interfaces and wireless communication. 8

Questions

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i. Identify appropriate wired and wireless communication protocols suitable for this wearable system and justify your choices.

ii. Compare SPI, I²C, and UART in terms of speed, implementation complexity, and power consumption when interfacing onboard sensors.

b) A smart utility meter installed in residential areas periodically records energy usage and transmits billing data to a central server using a cellular packet data connection.

Questions

i. Explain the sequence of steps required for the meter to establish a TCP/IP connection over the cellular data network to upload usage

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information.

ii. Describe the cellular communication architecture that enables reliable packet-based data transfer between the smart meter and the server.

6. a) A precision camera slider uses a stepper motor to move a mounted camera smoothly and accurately for time-lapse photography. The embedded controller must rotate the motor in fixed increments to achieve repeatable positioning. 7

Questions 8

i. Explain the operating principle of a stepper motor and discuss why it is well-suited for precise position control in embedded applications like this camera slider.

ii. The motor used has 400 steps per revolution. Calculate the step angle of the motor.

iii. If each step corresponds to 0.9°, determine how many steps are required to rotate the motor by 135°.

iv. During a motion sequence, the controller commands 600 steps. Calculate the total angular displacement of the motor.

b) A smart warehouse monitoring system tracks environmental conditions, equipment status, and inventory movement using connected embedded devices. Data from multiple sensor nodes is transmitted to a central monitoring platform for automation and alerts. 7

Questions

i. Describe suitable types of microcontrollers that could be used to implement this warehouse monitoring system and justify their selection.

ii. Identify the sensors and actuators required for monitoring environmental conditions and controlling warehouse equipment.

iii. Explain how the MQTT communication protocol supports efficient data exchange in this scenario, and describe the publish–subscribe model with the help of a diagram.

7. Write short notes on: (Any two) 2×5

a) Inter Process Communication

b) 7-segment display interfacing

c) IOT Communication models

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